From Company Websites to Official Product Codes: A Data-Driven Framework for PRODCOM Classification

Gabriele D'Amore and Donato Summa

Abstract This paper proposes an AI-driven framework to automate product classification in business surveys using corporate web data. Modular pipelines extract product information from company websites and assign official PRODCOM codes through unsupervised and supervised models. Acting as a digital assistant for statistical offices, the system reduces reporting burden and supports questionnaire prefilling. Results show strong alignment with official classifications, highlighting the potential of web-based AI tools as scalable support for product statistics.

Italiano. Questo articolo propone un framework di IA per automatizzare la classificazione dei prodotti nelle indagini sulle imprese usando dati web aziendali. Pipeline modulari estraggono prodotti dai siti e assegnano codici PRODCOM con modelli supervisionati e non. Il sistema riduce l'onere di risposta e supporta la precompilazione dei questionari. I risultati mostrano forte coerenza con le classificazioni ufficiali, evidenziando il potenziale di strumenti IA web scalabili per le statistiche sui prodotti.

Key words: AI, Product classification, Prodcom, Machine learning

1 Introduction

We address the bottleneck of product coding in business surveys, where heterogeneous product descriptions must be mapped to official classifications (e.g., PRODCOM). The task is time-consuming and error-prone because web and questionnaire texts are unstandardized, context-dependent, and often incomplete, while fine-grained codes differ by subtle semantic cues. We propose an AI-driven framework

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that uses corporate websites as auxiliary data for questionnaire pre-filling and coding validation, structured as modular pipelines that (i) identify product-relevant pages, (ii) extract name–description pairs, and (iii) assign codes via interchangeable engines: unsupervised embedding-based retrieval with LLM assisted classification and optional LLM disambiguation, or supervised models, a fine-tuned instruction LLM and a Joint Multi-task Variational Autoencoder (JMVAE). The system functions as a digital tool for statistical offices.

2 Problem Formalization and Broad Solution Strategy

Let U be the set of enterprises and Ω the set of potentially observable PRODCOM codes[1]. The inferential target is $A \subseteq U \times \Omega$, where $(i, \omega) \in A$ if enterprise i is associated with product code ω. The objective is to construct an approximation $\hat{A}^*$ of A from public web sources to support questionnaire pre-compilation and coding validation, consistently with recent methodological and experimental work on web-assisted product coding for PRODCOM-like surveys [2, 1, 10, 5].

Let Γ denote the set of accessible web pages and $X(h) = X_h$ the text extracted from page $h \in \Gamma$. For each enterprise i, let $\Gamma_i \subseteq \Gamma$ be its associated pages. In principle, the union of page-level evidence across Γ_i provides a web-based reconstruction of firm–product pairs; in practice, this reconstruction is incomplete because websites are selective representations of production and because some pages are unreachable or not detected as relevant. The automated solution is therefore expressed as a staged approximation. For each enterprise, the pipeline: (i) *URL Identification* identifies a filtered subset $\Gamma_i^* \subseteq \Gamma_i$ of relevant URLs, (ii) *Product Evidence Extraction* extracts from each $h \in \Gamma_i^*$ a reduced text $x_h^* \subseteq X_h$ containing product mentions, and (iii) *Code Assignment* assigns one or more predicted PRODCOM codes via an automatic classifier.

3 Methodology overview

The three stages introduced above—*URL Identification* (i), *Product Evidence Extraction* (ii), and *Code Assignment* (iii)—define a single modular methodology. We make explicit how (Γ_i, X_h, x_h^*) is transformed across stages and where alternative coding strategies differ.

We denote the stage mappings by

$$\Gamma_i \mapsto \Gamma_i^*, \qquad X_h \mapsto x_h^*, \qquad x_h^* \mapsto \widehat{\Omega}_{i,h},$$

where $\widehat{\Omega}_{i,h} \subseteq \Omega$ is the set of predicted codes obtained from page h for enterprise i.

[1] PRODCOM (*PRODuction COMmunautaire*) is the EU industrial production survey; PRODCOM codes are its official product identifiers (typically 8-digit).

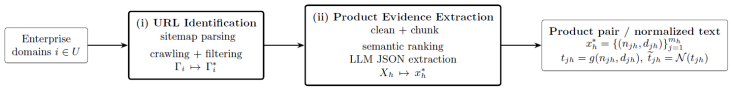

Fig. 1 The code assignment stage shares an upstream backbone.

(i) URL Identification. $\Gamma_i \mapsto \Gamma_i^*$

The first stage approximates the relevant evidence set by constructing $\Gamma_i^* \subseteq \Gamma_i$. Operationally, it combines domain discovery, `robots.txt` parsing, recursive sitemap expansion, URL inventory construction, and product-oriented filtering. In one reported large-scale crawling exercise, deterministic website discovery linked about 70% of enterprises to websites; recursive sitemap traversal produced more than 1.1 million URLs across 971 enterprises, later reduced to about 120,000 by product-oriented keyword filtering. An exploratory review of 174 websites identified recurrent patterns useful for filtering (direct product links in 42% of sites, downloadable catalogs in 7.5%, unavailable/incomplete sites in 14.9%), while confirming strong heterogeneity.

(ii) Product Evidence Extraction. $X_h \mapsto x_h^*$

Given a page $h \in \Gamma_i^*$ with text X_h, the second stage extracts a reduced representation $x_h^* \subseteq X_h$ containing product evidence. The procedure applies light pre-processing and selection of chunks from the text X_h, and then uses an LLM to output product records in JSON format. Candidate chunks are identified via retrieval-augmented generation (RAG). Formally,

$$x_h^* = \{(n_{jh}, d_{jh}) : j = 1, \ldots, m_h\},$$

where each pair contains a product name n_{jh} and a short description d_{jh}. In the open-source implementation, chunks are embedded with *snowflake-arctic-embed2* and ranked against the query embedding "Which products are listed?"; the top-ranked chunk is then passed to *Llama 3.1:8B* to return the JSON product list for the URL h. In a reported experiment on 26 companies, the extraction-plus-processing stage analyzed 2,614 URLs and yielded 1,083 extracted product records (duplicates retained to preserve contextual signals for stage (iii)).

(iii) Code Assignment.$x_h^* \mapsto \widehat{\Omega}_{i,h}$

The third stage maps extracted product pairs to one or more PRODCOM codes. For each $(n_{jh}, d_{jh}) \in x_h^*$ we rewrite the product text into a standardized, well-formatted input $\widetilde{t}_{jh}$ and predict a code $\hat{\omega}_{jh}$; the page-level output is $\widehat{\Omega}_{i,h} = \bigcup_{j=1}^{m_h} \{\hat{\omega}_{jh}\}$. Let

$$t_{jh} = g(n_{jh}, d_{jh}), \qquad \widetilde{t}_{jh} = \mathcal{N}(t_{jh}),$$

where g concatenates name and description and $\mathcal{N}$ denotes an optional LLM-assisted reformulation step via GPT-4o mini model. The coding stage for page h returns a candidate set $\widehat{\Omega}_{i,h} \subseteq \Omega$.

A key scoring function is cosine similarity. For embedding vectors $u, v \in \mathbb{R}^d$,

$$\cos(u,v) = \frac{u^\top v}{\|u\|\,\|v\|}.$$

In the proposed workflow, Cosine similarity is used twice: in stage (ii) to rank chunks against the product-query embedding, and in stage (iii) to rank PRODCOM candidates by comparing $e(\widetilde{t}_{jh})$ to official-description embeddings. The code-assignment stage supports two implementation families sharing the same interface but different decision rules.

Unsupervised retrieval branch: Let d_ω denote the official description associated with code $\omega \in \Omega$, and let $e(\cdot)$ be the embedding function computed using *text-embedding-3-large*; the similarity score of $\widetilde{t}_{jh}$ for code description d_ω is

$$s_{jh}(\omega) = \cos\big(e(\widetilde{t}_{jh}), e(d_\omega)\big),$$

with prediction obtained by nearest-neighbour search (e.g., in Qdrant [6]), optionally followed by LLM disambiguation over top-k candidates [8]. This branch minimizes labelling effort; the reported embedding-based route reached 45.00% exact eight-digit accuracy, with many non-exact predictions still matching higher-level taxonomy structure.

Supervised branch: two implementations share the same stage-(ii) output but differ in input representation and decision rule. *1) LABINLLAMA (fine-tuned LLM)*: the model exploits the normalized product text $\widetilde{t}_{jh}$ and outputs one PRODCOM code per extracted item. It is obtained by fine-tuning an open-source instruction model (Llama 3.2 3B Instruct) on labelled description–code pairs built from official nomenclature resources, namely CN8[2] product descriptions linked to PRODCOM codes through CN8–PRODCOM correspondence tables. At inference time, prompting with $\widetilde{t}_{jh}$ yields a single PRODCOM prediction, implementing a direct conditional mapping from product language to standardized labels. In the reported experiment, the upstream pipeline processed 2,614 URLs from 26 enterprises (complete page listings) and evaluated 89 extracted products against PRODCOM 2023 survey entries using the maximum number of sequential matching initial digits. All items achieved at least a 2-digit prefix match; the modal outcome was a 6-digit match; and more than 50% matched at least the first six digits, showing strong hierarchy awareness despite hallucinations. *2) JMVAE post-processing*: this classifier operates on the embedding $e(\widetilde{t}_{jh}) \in \mathbb{R}^d$ rather than on text. Following the VAE framework [3, 7], it learns an encoder $q_\phi(z \mid e(\widetilde{t}_{jh}))$ and a decoder $p_\theta(e(\widetilde{t}_{jh}) \mid z)$, so the VAE component regularizes the latent space via embedding reconstruction. Instead of a purely discriminative head, the network on z generates a proxy embedding intended to ap-

[2] CN8 (Combined Nomenclature 8-digit) is the European Union's tariff and statistical classification system for goods

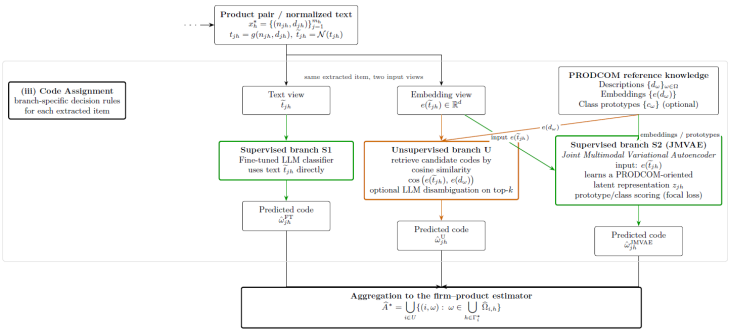

Fig. 2 Code assignment uses $\widetilde{t}_{jh}$ (LLM) or $e(\widetilde{t}_{jh})$ (retrieval/JMVAE) against PRODCOM embeddings and optional prototypes.

proximate the embedding of the PRODCOM description associated with the item; this addresses cases where cosine similarity between raw embeddings is unreliable due to style/context differences and embedding-model pre-training. Latent alignment is enforced via two decoders (one reconstructs the input embedding, the other produces a PRODCOM-oriented proxy vector) tied to the same latent distribution. The final prediction $\hat{\omega}_{jh}^{\text{JMVAE}}$ is obtained by cosine-similarity matching between the proxy vector and embeddings of PRODCOM descriptions (class vectors); cosine similarity is thus not the only modelling component, but it is the final matching criterion after latent alignment. The empirical experiment tackles 8-digit PRODCOM classification under severe class imbalance, mitigated via focal loss [4] and metric-learning objectives (cosine and triplet losses with hard-negative mining [9]). Since no product-level gold dataset is available, all train/validation/test splits are derived from the official CN8–PRODCOM conversion table, totaling 5,401 pairs spanning 2,378 PRODCOM codes (including 1,479 singletons). This setting measures mapping consistency and it does not substitute a product-level evaluation on web evidence.

4 Final Comparative Results

Table 1 summarizes evaluation metrics for the main explored configurations [1]. Three points emerge: (1) the unsupervised retrieval configuration (ChatGPT+Qdrant) provides a practical PRODCOM baseline (45.00% exact accuracy) without supervised training; (2) JMVAE post-processing improves exact eight-digit accuracy to 51.00%, showing that a learned latent representation over product-description embeddings recovers ambiguity left after stage-(ii). Weighted macro precision exceeds weighted macro recall (e.g., 71.27% vs 51.00% for JMVAE), suggesting that predictions are relatively accurate when made, while many true instances are missed (high false negatives), which is common under severe class imbalance. while unweighted

Table 1 Best empirical results.

Configuration	Set.	8-digit Accuracy	Macro-w. Precision	Macro-w. Recall	Macro-w. F1	Macro-unw. Precision	Macro-unw. Recall
ChatGPT + Qdrant	(i)	45.00%	–	–	–	–	–
JMVAE	(ii)	51.00%	71.27%	51.00%	69.84%	50.11%	47.51%

(i) Website→PRODCOM (extracted product evidence from company sites). *(ii)* CN8→PRODCOM (historical labeled CN8-description/PRODCOM pairs).Macro-weighted metrics are computed with R/*yardstick*, which excludes undefined per-class precision/F1 for labels with zero predicted support (TP+FP=0).

values (around 50%) expose the impact of severe imbalance and rare codes (many singletons), supporting focal loss and metric-learning components.

5 Conclusions and future work

The study shows that publicly available corporate web data can feasibly support the PRODCOM survey via modular, reproducible pipelines that streamline collection and product inventory building. Using crawling, text extraction, and both unsupervised and supervised models, predicted codes align well with PRODCOM at the 6–8 digit levels, though URL relevance detection and rare-code classification remain challenging. Near-term work focuses on refining embeddings and prompt design, especially to address cross-lingual differences in how pretrained models handle Italian versus English, while testing multimodal text+image models and fine-tuning strategies to reduce hallucinations (e.g., predicting the PRODCOM description associated with the correct code rather than the code itself) and implementing an LLM-based feedback loop for corrective refinement. Medium-term plans include deploying autonomous AI agents to gather additional web evidence, integrating complementary survey sources via Bayesian methods, and building a curated "golden" dataset.

References

1. D'Amore, G., Summa, D., Amarone, M. *et al.*: From web to code: AI-driven product classification for smarter business surveys. Paper ASA (2025)
2. D'Amore, G.: Methodological tools for automated extraction and classification of products for Prodcom survey. Working paper/manuscript (31 July 2025)
3. Kingma, D.P., Welling, M.: Auto-encoding variational Bayes. arXiv preprint arXiv:1312.6114 (2013)
4. Lin, T.-Y., Goyal, P., Girshick, R., He, K., Dollár, P.: Focal loss for dense object detection. In: Proceedings of the IEEE International Conference on Computer Vision, pp. 2980–2988 (2017)
5. Morrone, M., D'Amore, G., Summa, D. *et al.*: An experimental approach based on web intelligence and artificial intelligence techniques for the automatic classification of products. ENBES paper/manuscript (2025)
6. Qdrant: Vector database and vector similarity search engine. `https://qdrant.tech/`
7. Rezende, D.J., Mohamed, S., Wierstra, D.: Stochastic backpropagation and approximate inference in deep generative models. In: International Conference on Machine Learning, pp. 1278–1286. PMLR (2014)
8. Salton, G., Wong, A., Yang, C.S.: A vector space model for automatic indexing. Communications of the ACM **18**(11), 613–620 (1975)
9. Schroff, F., Kalenichenko, D., Philbin, J.: FaceNet: A unified embedding for face recognition and clustering. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 815–823 (2015)
10. Summa, D., D'Amore, G., Morrone, M.: Web-intelligence techniques for automatic product coding in economic surveys. UNECE CES Expert Meeting on Statistical Data Collection and Sources (Lisbon, 2025)